Table 6.2: notation of the leak codes

code	meaning	evidence
M	identical variant row in training (same protein family, identical mutation set, including reverse structures)	verified
m	every mutation in the row appears in training in the same protein, but not this combination (e.g. a test double whose two singles were trained)	verified
P	same protein (≥95 % identity over ≥90 % shorter-sequence coverage, or ≥90 % identity over ≥50 %) in training, no mutation match	verified
H	strong homolog in training: MMseqs2 E ≤ 1e-3 over ≥50 % of the test protein (plus Foldseek E ≤ 1e-3 or qTM > 0.70 over ≥50 % on Domainome and ProteinGym)	verified
h	weak/ambiguous: MMseqs2 E ≤ 1e-2 or ≥30 % identity over ≥30 % of the test protein (plus Foldseek E ≤ 1e-2 over ≥30 % on Domainome and ProteinGym)	measured
D	documented training on the same source dataset; rows not reconstructable	documented
–	no supervised ΔΔG training overlap (includes physics-based and unsupervised pretraining-only models)	verified / documented
U	training data not disclosed	documented
V	the dataset was used for validation or model selection	documented
#	the number is not a test result (the dataset is the training set)	verified
?	probable but unverified (e.g. M?: same protein and same source dataset, row-level file not public)	inferred
n.a.	not audited: no statement about overlap is made, and the cell is not eligible for bold	–

Table 6.3: training inventories

inventory	training data	source	evidence
ESM-MSR (released), ESM-MSR (devel), σ sweep, WT/MT heads	all substitutions of the 172 capped-train libraries (229,721 rows)	splits_oct06_capped.pkl + Tsuboyama2023_Dataset2_Dataset3_20230416.csv	verified
MutateEverything st1	the 172 capped-train libraries: 154,358 valid singles (150,815 wild-type-backbone + 3,543 mutant-background)	capped split; ME README recipe, stage 1 (20 epochs)	verified
MutateEverything st2	the same singles plus all 53,188 measured doubles (207,546 rows); stage 2 (100 epochs) initialised from the same seed's stage 1	capped split; ME README recipe	verified
ThermoMPNN (additive)	the same 154,358 singles	capped split; train_single.yaml	verified
ThermoMPNN-D	134 of the 172 capped-train libraries: the 53,188 measured doubles (52 libraries) plus 118,963 Rosetta-augmented doubles (82 additional natural libraries)	capped split; train_epistatic.yaml (05c_tmpnnd_inventory.py)	verified
SPURS (retrained)	the 172 capped-train libraries: 154,358 valid singles (150,815 wild-type-backbone + 3,543 mutant-background)	capped split; patched SPURS training code, three seeds	verified
SPURS multi (retrained)	the same singles (through its same-seed SPURS initialisation) and the 53,188 measured doubles	capped split; patched SPURS training code, three seeds	verified
EsmTherm	K50_dG_Dataset1_Dataset2 domains; 414 wildtypes contribute train mutants under a per-row 80/10/10 split; only 47 domains are test-only	EsmTherm/datasets/split/wildtype_split.csv	protein-level (mutation rows unavailable)
IFUM	Megascale Dataset2: 322 train libraries, 86 in an unseparated val/test block	IFUM published dataset_splits.pdf	protein-level
DDGemb	S2450 = S2648 − 18 sequences/198 mutations >25 % identity to S669; 2,450 mutations / 115 UniProt	ddgemb.biocomp.unibo.it S2450.tsv + S2450.fasta	verified (sequence-aligned)
DDMut	S9028 = S2648 + FireProtDB, 4,514 forward + 4,514 reverse; multi-mutant branch SM1242	biosig.lab.uq.edu.au S9028.csv, DD_SM1242.csv	verified (forward + reverse)
GeoDDG-3D	S8754 = 8,754 single mutations / 301 proteins (ProThermDB + ThermoMutDB)	GeoStab S8754.csv	verified (construct-aligned)
ESM3ΔG	960,216-sequence MGnify Stability fine-tune (525,127 WT + 435,089 point mutants); Megascale is its held-out test	Zenodo 19411306 mgnify_training_index.csv	verified (0 exact sequences)
ProSST	18.8 M AlphaFoldDB structures (masked-LM); no ΔΔG fine-tune (–)	ProSST paper	documented only
ProteinMPNN	PDB assemblies deposited before 2021-08-02; no mutation-level supervision (–)	ProteinMPNN paper	documented only
ESM3 (small/medium/large, and the σ = 0 control)	UniRef/MGnify/JGI + PDB pre-2020-05-01 + AFDB v4; no mutation-level supervision (–)	ESM3 paper	documented only
Rosetta Cartesian ΔΔG	no training data (–)	physics-based	

Table 6.5: what the audit finds

finding	size	affected rows
23 ProteinGym assays are fully inside the capped train split (+ 3 strong 3D structural homologs)	23/217 M + 3/217 H DMS, all Stability	45,623 M + 2 m + 4,607 H / 2,465,767 rows
ME stage 1/2, ThermoMPNN and retrained SPURS on those assays	singles (24,252 exact) + constituent (21,331) for ME/SPURS; 24,252 exact for TMPNN	same 23 M + 3 H DMS
Two Domainome domains (O95155_PF04564_1229 = 3L1X, Q9H161_PF00046_216 = 2M0C) match capped train at 100 % identity (+ 60 strong 2D/3D homolog domains)	2 M + 2 P + 60 H of 522 domains (458 low-id clean of M/P/H)	2,023 exact + 2,288 P (0.8 %) + 58,422 H rows (10.9 %); h.03 on the 458-domain subset
EsmTherm's 80/10/10 is per row, not per domain: 27/33 test libraries (including mutant backgrounds) contribute training mutants	85.6 % of test rows same-protein (P.86)	P, M? expected ≈ 0.8 (M?.68)
IFUM trained on 21/33 test libraries (including mutant backgrounds)	48.6 % of test rows same-protein (P.49)	P, M? = same rows (M?.49)
ESM3ΔG is zero-shot on Megascale	0 exact sequences anywhere in the benchmark	40.7 % strong (8/33 libs) + 50.3 % weak (21/33) homologs
DDGemb = S2450 (sequence-aligned S2450.tsv + S2450.fasta)	92.5 % of S2648; 63.4 % of Q3421; 43.6 % of S8754; 38.3 % of K3822; 33.0 % of K2369; 29.7 % of Ssym; m.48/m.54 on PTMUL/PTMUL-D; 0 % on S783, S669, S461	verified exact / constituent
DDMut = S9028 + SM1242 (forward + reverse structures)	98.2 % of S2648; 88.3 % of Ssym; 83.5 % of Q3421; 74.3 % of K2369; 72.2 % of K3822; 69.5 % of S8754; 60.9 % of S783; 58.9 %/71.1 % of PTMUL/PTMUL-D; 10.0 %/14.5 % of S669/S461	verified exact
GeoDDG = S8754 (construct-aligned)	S8754 is its training set (#); 100 % of S2648; 97.9 % of Q3421; 91.2 % of K2369; 89.1 % of K3822; 88.4 % of S783; 50.0 % of Ssym; m.73/m.78 on PTMUL/PTMUL-D; 1.2 % of S669	verified exact / constituent
the "six clean sets" claim	only S461 has zero identical mutations across all full-coverage models	DDMut has M.10/M.15 on S669/S461; GeoDDG has M.01 H.03 on S669